Supplementary file2 **Meta-regression analysis**

1.Purpose: Meta-regression analysis was performed to analyze the potential heterogeneity.

2.Mehtod: **Meta-regression analysis**

We used a bivariat random effect meta-regression to explore potential sources of heterogeneity. The following covariates were used in the meta-regression analysis: (1) grading system (FGS vs WHO/ISUP grading system); (2) Segmentation software (ITK-SNAP vs others).

3.Results of meta-regression analysis

Meta regression was performed on the two covariates (grading systems and segmentation software) as 11 studies were included. Results revealed that grading systems and segmentation software were not a significant association with the heterogeneity. Specific results of the meta-regression analysis are summarized in the following **Table3.**

Table 3. Results of meta-regression analysis

Covariate	Studies (n)	Sensitivity (95% CI)	Specitivity (95% CI)	P value
Grading system				
Fuhrman	8	0.76 (0.72-0.79)	0.83 (0.79-0.87)	0.39
WHO/ISUP	3	0.82 (0.77-0.86)	0.85 (0.79-0.90)	
Segmentation software				
ITK-SNAP	4	0.77 (0.74-0.81)	0.84 (0.80-0.89)	0.83
Others	7	0.77 (0.73-0.81)	0.83 (0.79-0.87)	